Storage Lab: Unresponsive Mount Clean Teardown & Lazy Unmount Recovery

Duration : 30 Min.

Scenario

A remote NFS server crashed, leaving multiple Linux client processes hanging in 'D' (uninterruptible sleep) state while attempting to read from /mnt/remote_nfs. Standard 'umount /mnt/remote_nfs' hangs. The junior engineer is tempted to reboot the server, but other business-critical containers are running on the host. The engineer must execute a clean teardown of the dead mount using fuser and umount -l.

Task Objectives

Perform the following unresponsive mount recovery tasks:

1. Detect that the remote mount point `/mnt/remote_nfs` is unresponsive using safe timeout detection techniques.
2. Identify all worker processes trapped in uninterruptible sleep or holding descriptors on the dead share.
3. Terminate blocking processes cleanly and execute a safe lazy detachment of the hung filesystem without server reboot.
4. Confirm that the system mount table is consistent and that storage utilities operate normally without hanging.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. If configuration files, backup manifests, or diagnostic reports are required, save them inside the `student_workspace` directory.
4. Verify your storage configurations locally by executing: `bash python run.py`
5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Mount hung detection: timeout command detects mount freeze without causing diagnostic tool lockup	3 Marks
TC2	Process termination: fuser cleanly signals processes in D-state holding descriptors to the dead share	2 Marks
TC3	Lazy unmount execution: umount -l successfully detaches the unresponsive filesystem from directory tree	3 Marks
TC4	State verification: /proc/mounts confirms removal and df operates normally without hanging	2 Marks

Total Score: 10 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.